

Problem #1a: Accessing public exoplanet data (5 pts)

The largest catalog of exoplanets can be found in the [Composite Data Table](#) of the NASA Exoplanet Archive. Download the data table by clicking on the "Download" button, and save to a datafile that you can open locally on your computer/laptop. To verify that you have accessed the data, plot a histogram of the log-base-10 of the orbital period of each exoplanet. Be sure to label your axes!

Hint: there are several header columns in the data file that will be downloaded. These headers show you the names of the columns (including the name of the orbital period column).

Hint 2: Pandas makes loading data easy. You can specify the number of header lines to skip using 'skiprows=number_to_skip' in the function call that loads the data. If you skip the right number of rows, you can use the row just before the data starts to specify the columns automatically!

Problem #1b: Multiplanet systems (10 pts)

Now that you have data loaded, you should use the column which specifies the number of planets per system to plot histograms for subsets of the data containing 1, 2, 3, 4, and 5 planets per system on the same figure. Be sure to include a legend so that you can see how the orbital period distributions change.

Problem #1c: Biases in observing multiplanet systems (9 pts)

Based on your figure and the material in the class lectures, provide two examples of how observational biases could affect the size and orbital period distribution of planetary systems. Provide one argument, based on exoplanet dynamics, for why there could truly be fewer multi-planet systems.

Problem #2: Figures and captions (again!) (20 points)

Your project presentations are coming up soon! As part of your presentation, you should plan to include at least one original figure that you (and your partner if you have one) make. In the presentation, you'll be able to walk the audience through the figure. This isn't the only way that figures are presented though!

For this problem, you should create a figure (you and your partner can submit the same figure) that you could use in your presentation. The figure should illustrate some conclusion or result from your project and should have adequate axis labels and a legend if necessary. Once you have a figure, you should also write a figure caption that describes the necessary information for a reader to understand and interpret the figure themselves. It's also a good idea to add in a punchline that you want your reader to take away from the figure. *You must write your own figure caption even if you and your partner share the figure.*